common people. Hence people prefer low cost fuel for the vehicles. Diesel is a low cost fuel than petrol and it has high thermal energy and pulling power. In this way when diesel is used as fuel the method of supplying the fuel and combusting it becomes different from using petrol in Otto cycle. In the diesel cycle based engines a separate fuel supply system is used for introducing fuel inside the combustion chamber of the engine. The fuel supply system of the diesel engine has the following components.

1. Fuel Tank
2. Primary (course) Fuel Filter
3. Fuel feed pump
4. Secondary (fine) fuel filter
5. Fuel Injection Pump
6. Fuel Injector or Nozzle

In the above components of the fuel supply system the fuel tank, fuel filter, fuel feed pump were already discussed in detail. Hence here the details of types of fuel injection pumps and their arrangements can be seen in detail.

9.9 FUEL INJECTION PUMPS

In any diesel engine for producing required power, specified amount of fuel at the specified pressure and at the specified time must be introduced inside the combustion chamber. The device used for achieving such operation is called as the fuel injection pump. If the fuel is injected in such high pressure inside combustion chamber which has the compressed air at very high temperature then the fuel will get auto ignited and the engine power will increase. The pumps used for such operation are classified into the following types.

1. In line or jerk type Pumps
2. Distributor pumps

9.9.1 Inline (Jerk type) Pump

When the fuel is supplied through the intake manifold by induction like carburetion due to wall wetting effect appropriate fuel will not be supplied to the individual cylinders. Due to this improper distribution of the fuel supply, individual cylinders produce varying power outputs in multi cylinder engines. This distribution of the fuel could be avoided by injecting the fuel by using separate injectors fitted at the individual cylinders of the engine. In multi cylinder engines when the appropriate amount of fuel is injected at the appropriate timings as per the firing order then the power variations can be eliminated. In multi cylinder engines the injectors are mounted at each cylinder of the engine and arranged in line and operated by individual injection pumps which are operated by a common cam shaft. These types of pumps are called as in line pumps. The injection pressure in such pumps can be increased from 7 Mpa (mega pascal) to 30 Mpa. In a diesel engine operating at 6000 rpm with 150 mm^3 quantity of the fuel will be injected 20 times. The In line pumps can be also called as jerk pumps.

Construction:

In an in line diesel pump plunger, barrel and delivery unit are the important components present. The plunger is kept inside the barrel in correct fit and allowed to move inside the plunger barrel up and down. Tooth quadrant provided on the barrel

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